
MA 705: Modelling Population Dynamics

Course Code	MA 705
Course Name	Modelling Population Dynamics
L-T-P-C	3-0-0-3
One-time approval	26th BoA (Oct 2018); 18th Senate (Oct 2018)
Recoded to	MA 604 by the 30th BoA (Jun 2019)

Course Description

This course is an introduction to synthesis of mathematical models and empirical ecology. Why do organisms become extremely abundant one year, and then apparently disappear a few years later? Why population outbreaks in certain species in certain locations happen more or less regularly, while in other locations such eruptions are irregular? This course will provide answers to such questions. From this course, students will gain a better understanding of population dynamics in general.

Course Contents

- **Module 1:** Modeling Nature, General philosophy of the approach.
- **Module 2:** Population Dynamics from the first principle, Exponential Growth, Self Limitation, Consumer Resource Oscillation.
- **Module 3:** Types of Models: Continuous-time, Discrete-time, Delayed Differential models, Exogenous Drivers, Age and Stage- structured Models.
- **Module 4:** Population Interactions: Functional response, Aggregative response, Numerical Response, Lotka Volterra Model, Anatomy of Predator Prey Cycle, Grazing systems, pathogens and parasites, Tritrophic Models.
- **Module 5:** Empirical Approaches: Analysis of Population fluctuations, Time Series Analysis, Fitting Models to Data, Fitting Mechanistic Models.